NVIDIA Stock Market Analysis

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1. Project Objectives

The objective of this project is to explore the patterns, trends, and characteristics of NVIDIA (NVDA) stock performance using historical stock market data.

This analysis focuses on stock prices, trading volume, daily returns, and long-term trends in order to identify meaningful insights about NVIDIA's market performance.

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2. Dataset Description

The dataset used in this project is historical stock market data for NVIDIA Corporation (NVDA), obtained from Yahoo Finance.

The CSV file contains the following columns:

- Date: Trading date
- Open: Opening stock price
- High: Highest stock price of the day
- Low: Lowest stock price of the day
- Close: Closing stock price
- Volume: Number of shares traded

The dataset was downloaded as a CSV file and loaded into Python using pandas.

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3. Pseudocode

Below is the pseudocode used to guide this project:

1. Load NVDA stock dataset
2. Remove missing or invalid values
3. Convert Date column into proper datetime format

4. Extract year from Date for yearly trend analysis
5. Generate descriptive statistics for stock prices and trading volume
6. Create visualizations:
    - Closing price trend line chart
    - Trading volume bar chart
    - Daily return distribution histogram
    - Moving average analysis chart
7. Summarize key findings and insights

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4. Algorithm Implemented

A simple algorithm was designed to analyze long-term stock price trends of NVIDIA.

Algorithm: Yearly Average Closing Price

Input: Dataset with Date and Close columns

Process:

1. Convert Date to pandas datetime type
2. Extract the "year" component
3. Group stock prices by year
4. Calculate average closing price for each year

Output:

A dictionary-like structure mapping each year to its average stock price.

This algorithm is used to evaluate whether NVIDIA stock shows long-term growth or decline over time.

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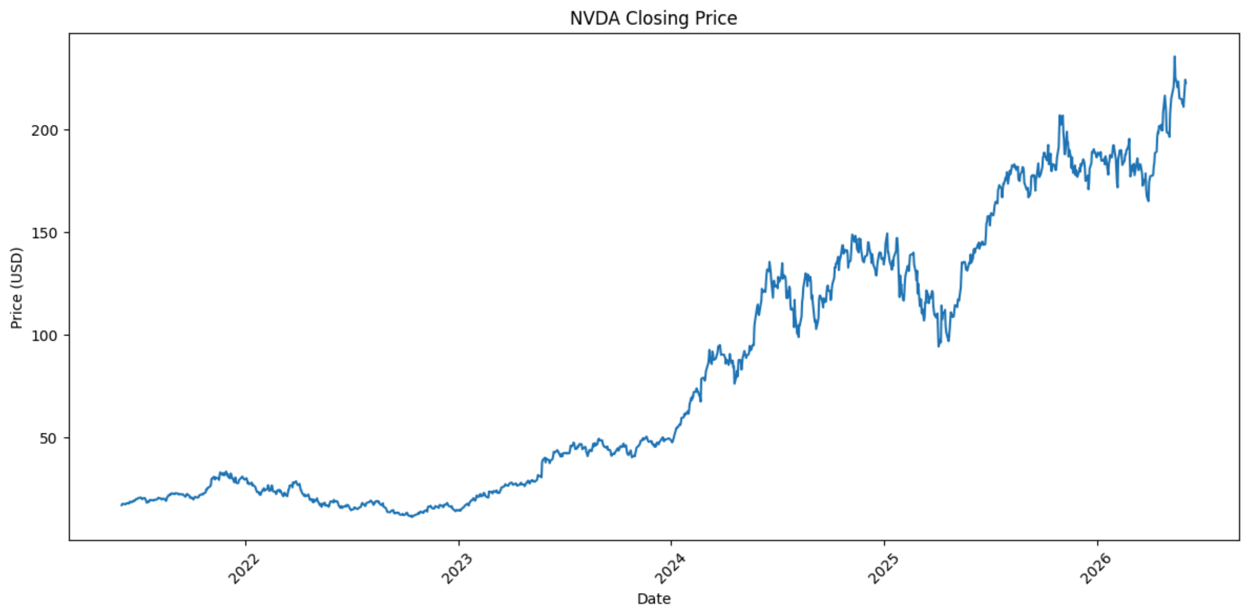
5. Exploratory Data Analysis (EDA)



5.1 Descriptive Statistics

- NVIDIA stock price has increased significantly over recent years.
- Trading volume fluctuates according to market activity.
- Stock returns show periods of both high growth and volatility.

5.2 Visualization 1 — Closing Price Trend

A line chart was generated to illustrate the historical closing price of NVIDIA stock.

The chart clearly shows a strong upward trend over the long term, especially during the AI industry boom.

5.3 Visualization 2 — Trading Volume

A bar chart was created to examine trading activity.

Periods with unusually high trading volume often correspond to important company announcements or major market events.

5.4 Visualization 3 — Daily Return Distribution

A histogram was produced to visualize the distribution of daily returns.

Most daily returns are concentrated around zero, indicating that large price changes occur relatively infrequently.

5.5 Visualization 4 — Moving Average Analysis

A moving average chart was created using the closing prices.

The moving average helps smooth short-term fluctuations and provides a clearer view of long-term trends.

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6. Summary of Findings

- NVIDIA stock has demonstrated strong long-term growth.
- Trading volume increases during major market events.
- Most daily stock returns are relatively small.
- Moving averages effectively reveal long-term trends.
- NVIDIA's stock performance reflects the rapid growth of the AI and semiconductor industries.

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7. Conclusion

This project successfully demonstrates stock market data analysis using Python, including data cleaning, descriptive statistics, algorithm implementation, and multiple visualizations.

The results provide valuable insights into NVIDIA's historical stock performance and market behavior.

GitHub Link

GitHub Repository: